

- 5 Yeast cells are unicellular eukaryotes. Yeast cells respond to changes in the concentrations of the sugars galactose and glucose by switching on genes using transcription factors.

Fig. 5.1 shows the events that occur when the sugar galactose is present and glucose is absent in the external environment.

Msn2 transcription factor proteins **bind** to promoters of genes *A* and *B*

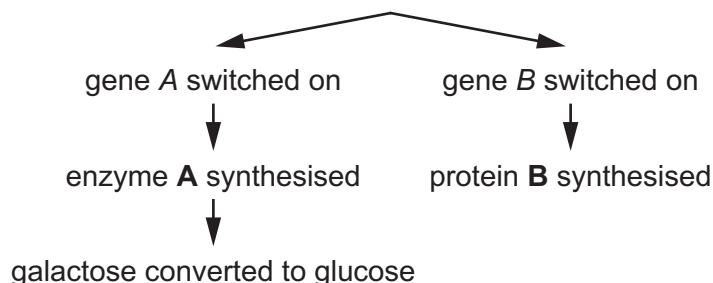


Fig. 5.1

- (a) (i) Explain why enzyme **A** is described as an inducible enzyme.

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..... [3]

- (ii) Protein **B** has a similar role to the protein coded for by gene *Y* of the *lac* operon.

Suggest a possible role for protein **B** that would allow yeast cells to make use of galactose.

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..... [1]

Scientists produced genetically engineered yeast cells. The gene coding for Msn2 transcription factor proteins and the marker gene coding for green fluorescent protein (GFP) are transcribed together to produce a single mRNA molecule.

The resulting Msn2 transcription factor proteins contain a GFP region as part of their structure and are called tagged Msn2 molecules. These tagged Msn2 molecules show up as green fluorescent spots when viewed using a microscope with a very high resolution.

An investigation was carried out to compare the distribution of tagged Msn2 molecules in yeast cells, when glucose is absent and when glucose is present.

The results are shown in Table 5.1.

Table 5.1

glucose availability	mean number of tagged Msn2 molecules present		
	cytoplasm	nucleus	total
glucose absent	3 263	2 012	5 275
glucose present	1 755	632	2 387

- (b) (i) Calculate the percentage of Msn2 molecules in the cell that are located inside the nucleus, when glucose is **absent**.

Show your working and write your answer to **two** significant figures.

..... % [2]

- (ii) When glucose is **present**, 26% of Msn2 molecules in the cell are located inside the nucleus.

Suggest why this figure is different from your answer to (i).

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 [2]

- (c) Another method of tagging molecules with a fluorescent colour is to use monoclonal antibodies that have GFP attached. These monoclonal antibodies bind specifically to the molecule of interest. This is called immunofluorescent tagging.

Suggest reasons why immunofluorescent tagging is **not** a suitable choice for this investigation.

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..... [3]

[Total: 11]